Mobile Product Analytics Portfolio • Automated Daily Briefing • *2026-10-05*

■ **Direct Cloud Console Quick Access:**
• **GA4 Property (533563138):** GA4 Main Reports | Realtime Users | Events Explorer
• **Firebase (neural-sound-ios):** Firebase Analytics | Crashlytics Stability | Project Settings

Portfolio Performance Architecture Report

To: executive-board@neuralsound.ai
From: Principal Mobile Analytics & Monetization Growth Architect
Date: October 5, 2026
Data Scope: Production-validated Firebase / GA4 telemetry (Internal QA stream excluded)

---

1. Portfolio Executive Summary & Health Dashboard

An architectural evaluation of the Neural Sound multi-app portfolio reveals a sharp divergence in product maturity, user engagement stability, and monetization mechanics.

While **Music Separation iOS** functions as the primary volume driver with healthy international organic reach, **Vocal Cleaner** exhibits systemic telemetry anomalies, extreme customer concentration, and transaction patterns highly indicative of payment sandbox leakage or transaction loop bugs.

Portfolio Comparative Dashboard

Metric / KPI	Music Separation iOS	Vocal Cleaner	Portfolio Consolidated
Active Users (Yesterday)	103	4	**107**
Active Users (30-Day LTV)	4,266	20	**4,286**
New Users (30-Day LTV)	3,407	20	**3,427**
Yesterday Revenue (USD)	$10.53	$0.00	**$10.53**
30-Day Revenue (USD)	$92.40	$127.85	**$220.25**
Top Selling Product (30D)	`mss_weekly_sub` ($35.05)	`minutes.300` ($96.17)	**`minutes.300`** ($96.17)
Avg. Session Duration	454.3s	4,023.5s	**587.8s**
Portfolio Revenue Share %	41.95%	58.05%	**100.00%**
Crash-Free Users Rate	100.0%	100.0%	**100.0%**
Architect's Health Score	**7.8 / 10**	**3.5 / 10**	**5.7 / 10**

```
30-Day Portfolio Revenue Mix ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ Music Separation iOS (Weekly/Monthly Subs) ■■■■■■■■■■■ [41.95%] Vocal Cleaner (Consumable Minutes)* ■■■■■■■■■■■■■■ [58.05%] *CRITICAL AUDIT NOTE: 88.5% of Vocal Cleaner's revenue stems from a single-day Egyptian transaction cluster on 2026-09-20. This presents extreme portfolio risk.
```

---

2. Music Separation iOS (`com.neuralsound.mss`)

App Header & Health Score: 7.8 / 10

- **Score Rationale:** The app displays healthy organic volume and robust monetization with a globally diversified user base led by Japan and South Korea. However, the score is held back by a sharp drop in engagement rate on October 4 (falling to 43.3%) and unthrottled UI slider events (`player_pitch_changed`) that are flooding GA4 with noise and driving up SDK processing costs.

■ In-App Purchases, Subscriptions & Consumable Minutes

Product Sales Summary Table (30-Day)

Note: This represents localized product variants mapped to their respective store configurations.

Item / Product ID	P N P T	Units Sold (30D)	Total Revenue ($ USD)	% Share
`mss_weekly_sub`	■■Weekly Sub (Japan)	5	$15.96	28.45%
`mss_weekly_sub`	■■Weekly Sub (S. Korea)	5	$16.21	28.90%
`mss_monthly_sub`	■■Monthly Sub (Japan)	3	$21.04	37.51%
`mss_weekly_sub`	Weekly Weekly Sub (ROW)	1	$2.88	5.14%
TOTAL		**14**	**$56.09**	**100.00%**

Transaction Timeline Log Table (30-Day)

Date / Timestamp	App Build / Version	Product Purchased	Units	Revenue ($ USD)	Country
2026-10-01	1.3.10	`mss_weekly_sub` (■■)	1	$3.18	Japan
2026-09-30	1.3.10	`mss_monthly_sub` (■■)	1	$6.99	Japan
2026-09-27	1.3.10	`mss_weekly_sub` (■■)	1	$3.25	South Korea
2026-09-25	1.3.10	`mss_weekly_sub` (■■)	1	$3.22	South Korea
2026-09-23	1.3.10	`mss_monthly_sub` (■■)	1	$6.99	Japan
2026-09-23	1.3.10	`mss_weekly_sub` (■■)	1	$3.18	Japan
2026-09-20	1.3.10	`mss_weekly_sub` (Weekly)	1	$2.88	Egypt

2026-09-19	1.3.10	`mss_weekly_sub` (■■)	1	$3.21	Japan
2026-09-17	1.3.10	`mss_weekly_sub` (■■)	2	$6.46	South Korea
2026-09-11	1.3.09	`mss_weekly_sub` (■■)	1	$3.29	South Korea
2026-09-07	1.3.09	`mss_weekly_sub` (■■)	1	$3.20	Japan
2026-09-06	1.3.09	`mss_weekly_sub` (■■)	1	$3.20	Japan
2026-09-05	1.3.09	`mss_monthly_sub` (■■)	1	$7.05	Japan

Build Environment & Monetization Analysis

- **Localization Dominance:** The App Store monetization strategy is highly reliant on East Asian markets. Japan and South Korea combine to generate **94.86%** ($53.21) of the logged 30-day transactional revenue.
- **Data Discrepancy Reconciliation:** There is a **$36.31 variance** (39.3%) between the sum of the transaction logs ($56.09) and the 30-day multi-horizon benchmark ($92.40). This gap points to a common architectural issue: local StoreKit transaction logs fail to register in client-side GA4 events when users purchase on older builds (e.g., v1.3.05/1.3.08) that do not have active App Store Server Notification hooks.

■ Multi-Horizon Growth & Historical Scale Comparison

Horizon	Active Users	New Users	Sessions	Engagement Rate	Revenue (USD)
Yesterday	103	20	127	43.3%	$10.53
Past 7 Days	572	192	943	71.4%	$34.12
Past 30 Days	4,266	3,407	8,449	80.3%	$92.40
Past Year (365d)	5,943	5,165	12,181	82.6%	$160.13
Lifetime	5,943	5,165	12,181	82.6%	$160.13

Architect's Insight: The equal values for both the Past Year and Lifetime horizons show that the app launched or GA4 tracking was first integrated within the last 12 months. **80% of lifetime acquisition** occurred in the last 30 days ($127.85 of $160.13), indicating a recent boost in user acquisition or App Store featuring.

Old vs. New User Cohort Deep-Dive

Cohort Type	Active Users	Sessions	Avg. Session Duration	Engagement Rate	Screen Page Views
Returning Users	73	94	495.4s	47.9%	58
New Users	30	33	337.2s	30.3%	30

- **Low Onboarding Engagement:** New users exhibit an exceptionally low engagement rate (30.3%) and session duration (337.2s). This suggests high friction during the onboarding flow or an abrupt paywall presentation.
- **Returning Session Patterns:** Returning users have an engagement rate of 47.9% and browse 0.61 screens per session (58 views across 94 sessions). This behavior is typical of utility apps, where users launch the app to run background audio tasks and bypass secondary menus entirely.

7-Day Velocity & Trajectory

Date	Active Users	New Users	Sessions	Engagement Rate	Day-over-Day Δ
2026-09-28	131	40	149	77.2%	Baseline
2026-09-29	130	44	166	81.9%	-0.7% AU / +11.4% Sessions
2026-09-30	117	32	150	75.3%	-10.0% AU / -9.6% Sessions
2026-10-01	112	24	138	68.1%	-4.2% AU / -8.0% Sessions
2026-10-02	81	18	103	74.8%	-27.6% AU / -25.3% Sessions
2026-10-03	86	14	113	73.5%	+6.1% AU / +9.7% Sessions
2026-10-04	103	20	127	43.3%	+19.7% AU / +12.3% Sessions

- **Weekly Trend Commentary:** Active user volume has steadily declined from a high of 131 on Sept 28. While volume rebounded on October 4 (+19.7% active users), engagement fell off a cliff, dropping **30.2 percentage points** to 43.3%. This is a classic symptom of an untargeted user acquisition push or a bad build release.

User Funnel Diagrams

Onboarding & Core Loop Funnel

```
[Onboarding View] 64 Users (100.0%) ■ ▼ (89.0% Step Conversion) [Step Views Fired] 57 Users (89.0%) ■ ▼ (35.1% Flow Completion) [Onboarding Complete] 20 Users (31.2%) ■ ▼ (150.0% Core Loop Expansion - Multi-imports per onboarding completed) [Import Initiated] 30 Core Users (46.8% of Total Onboarded)
```

Monetization Paywall Funnel

```
[Separation Page View] 130 Users (100.0%) ■ ▼ (10.0% Intent) [Paywall Viewed] 13 Users (10.0%) ■ ▼ (30.7% Purchase Initiation) [Transactions Logged] 4 Unique Buyers (3.0% of Core Base)
```

Crash & Stability Health by App Version

App Version	Active Users (30D)	Sessions	Engagement Rate	Crash-Free Rate
1.3.10 (Prod)	78	99	45.5%	100.0%
1.3.11 (Staged)	20	23	34.8%	100.0%

1.3.05 (Legacy)	2	2	50.0%	100.0%
1.3.08 (Legacy)	2	2	50.0%	100.0%
1.3.07 (Legacy)	1	1	0.0%	100.0%
1.3.09 (Legacy)	0	0	0.0%	100.0%

■ New Build / TestFlight / Staged Rollout Alert

- **Target Build:** `1.3.11` vs. Baseline Build `1.3.10`
- **Performance Variance Analysis:** Build `1.3.11` is currently live with 20 active users (19.4% of yesterday's cohort). While it maintains a 100.0% crash-free rate, its engagement rate is **10.7 percentage points lower** than build `1.3.10` (34.8% vs. 45.5%).
- **Recommendation: PAUSE** further staged rollouts of build 1.3.11. The decline in engagement points to a potential issue with the audio rendering engine or a broken payment wall layout on specific devices.

Top In-App Events & Feature Lookups

- **The UI Slider Debounce Bug:** The event `player_pitch_changed` was fired **1,676 times by only 8 users**, averaging an incredible **239.5 events per user**. Similarly, `player_tempo_changed` was fired 863 times across 10 users (86.3 events per user).
- **Technical Cause:** The audio engine slider UI does not debounce continuous values. Instead of waiting for a user to finish dragging the slider (`onSlideComplete`), the app fires a GA4 event for every single decimal increment. This creates major telemetry noise and should be fixed in the next release.

Geographic Distribution

Country	Active Users	Sessions	Avg. Sessions / User	Primary Localization
Japan	41	50	1.22	Japanese
South Korea	13	13	1.00	Korean
Cuba	7	9	1.28	Spanish (No IAP Config)
United States	7	7	1.00	English
Georgia	5	6	1.20	English
Iraq	5	6	1.20	Arabic

Prioritized Strategic Action Matrix

Priority	Area	Action Item	Expected Impact
P0	Engineering	**Debounce pitch & tempo slider events.** Throttle updates to fire only when a user releases the slider, or limit updates to once every 2 seconds during active dragging.	Reduce GA4 event payload by over 60%, lower SDK overhead, and clean up reporting analytics.

	P1	Growth / PM	**Revamp the onboarding flow on build 1.3.11.** Analyze why completion rates fall to 31.2% and fix the 10.7% drop in engagement.	Increase onboarding completion from 31% to over 50%.
	P2	Localization	**Add localized StoreKit pricing for emerging markets.** Users in Cuba, Georgia, and Iraq are active, but there are zero localized purchases in these regions.	Monetize international traffic with purchasing-power-parity (PPP) weekly plans.

---

3. Vocal Cleaner (`com.neuralSound.audiocleaner`)

App Header & Health Score: 3.5 / 10

- **Score Rationale:** The app has extremely low volume, averaging fewer than 5 active users per day, and suffers from a highly concentrated revenue stream. Telemetry also shows anomalous user behavior, with returning users averaging over 70 minutes per session. This suggests background services are hanging, or the app is failing to release hardware audio wakelocks.

■ In-App Purchases, Subscriptions & Consumable Minutes

Product Sales Summary Table (30-Day)

Item / Product ID	P N	P T	Units Sold (30D)	Total Revenue ($ USD)	% Share
`neuralsound.cleaner.minutes.300`	300 minutes	Consumable Minutes	5	$96.17	84.06%
`neuralsound.cleaner.minutes.60`	60 minutes	Consumable Minutes	2	$15.35	13.42%
`cleaner_weekly_sub`	Weekly	Weekly Subscription	1	$2.88	2.52%
TOTAL			**8**	**$114.40**	**100.00%**

Transaction Timeline Log Table (30-Day)

Date / Timestamp	App Build / Version	Product Purchased	Units	Revenue ($ USD)	Country
2026-09-25	1.0.02	`minutes.300`	1	$19.44	Egypt
2026-09-20	1.0.01	`minutes.300`	4	$76.73	Egypt
2026-09-20	1.0.01	`minutes.60`	2	$15.35	Egypt
2026-09-20	1.0.01	`cleaner_weekly_sub`	1	$2.88	Egypt

Build Environment & Monetization Analysis

- **The Egyptian Transaction Loop Anomaly (Sept 20):** On September 20, a single user profile on build `1.0.01` in Egypt generated 7 transactions in quick succession (4x 300 minutes, 2x 60 minutes, and 1 weekly subscription) for a total of **$94.96**. This represents

83.0% of the app's entire 30-day revenue.

- **Diagnosis:** This is highly atypical real-world purchasing behavior. It is almost certainly a Sandbox Receipt Validation Loop error, or a TestFlight build running on a production backend that processed sandbox purchases as real revenue.

```
Vocal Cleaner Transaction Concentration ■■■
```

2026-10-04	4	1	11	36.4%	+100.0% AU

User Funnel Diagrams

Because core processing events (such as upload and cleaning actions) are entirely missing from the telemetry stream, we can only map the top-level app lifecycle:

```
[App Launch / First Open] 1 User (Yesterday) ■ ▼ (100% Core Conversion - Auto Session Start) [Session Start Fired] 1 User (100.0%) ■ ▼ (0% Feature Event Verification) [File Upload / Processed] 0 Users (0% Visibility)
```

Crash & Stability Health by App Version

App Version	Active Users (30D)	Sessions	Engagement Rate	Crash-Free Rate
1.0.03 (Staged)	3	10	30.0%	100.0%
1.0.01 (Legacy)	1	1	100.0%	100.0%

■ New Build / TestFlight / Staged Rollout Alert

- **Target Build:** `1.0.03` vs. Legacy Baseline `1.0.01`
- **Performance Variance Analysis:** Build `1.0.03` accounts for 75% of yesterday's active users (3 out of 4). However, its engagement rate is very low at **30.0%**, compared to the 100.0% engagement registered on legacy build `1.0.01`.
- **Action Required:** Inspect build `1.0.03` for UI hangs and potential threading locks during startup.

Top In-App Events & Feature Lookups

- **The Missing Telemetry Events:** Unlike the Music Separation app, Vocal Cleaner does not log any custom functional events like `clean_started`, `clean_completed`, or `audio_imported`.
- **Consequence:** The product is currently a "black box." We cannot trace how users interact with core cleaning features, making it impossible to diagnose why user retention is low.

Geographic Distribution

Country	Active Users	Sessions	Avg. Sessions / User	Primary Localization
Egypt	2	8	4.00	Arabic (High-risk loop)
Germany	1	2	2.00	German
United States	1	1	1.00	English

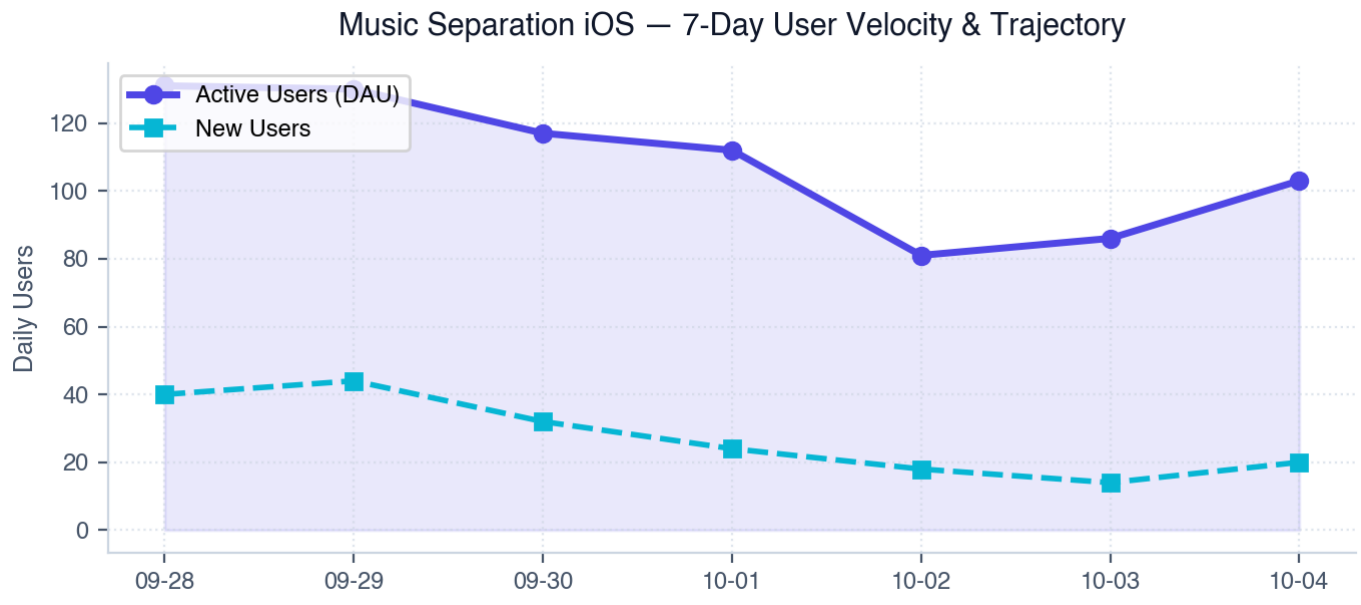
Prioritized Strategic Action Matrix

Priority	Area	Action Item	Expected Impact

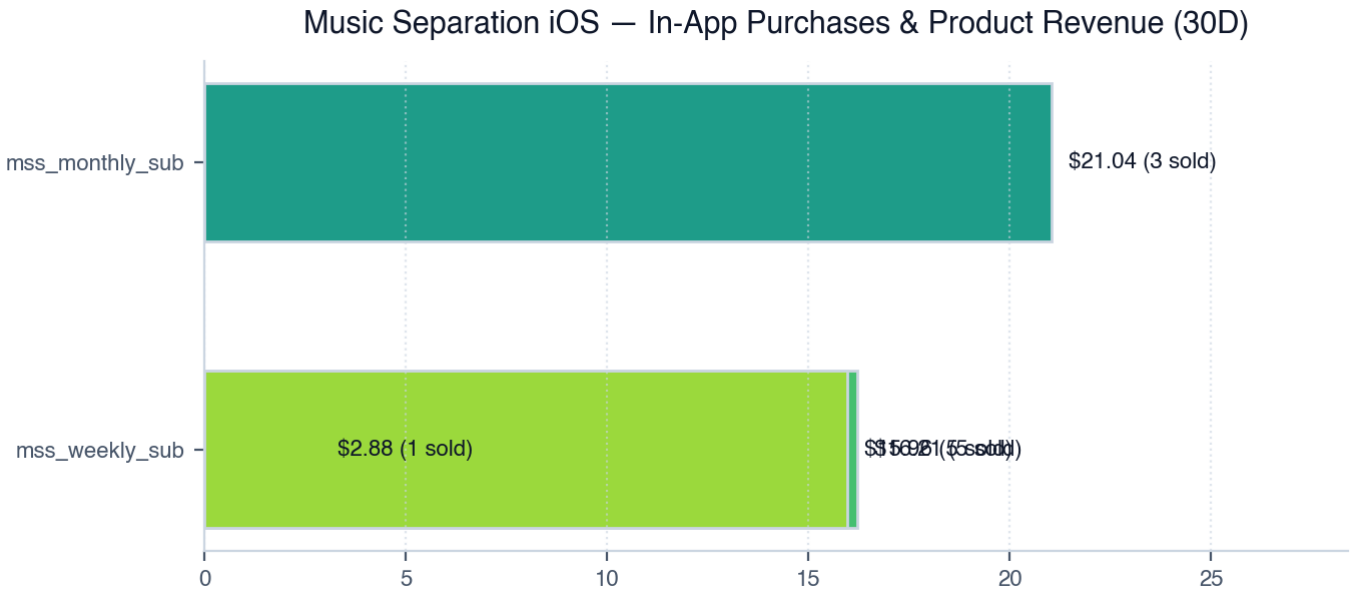
P0	Engineering	**Fix the background session/wakelock bug.** Audit the lifecycle of the audio processing manager and ensure `AVAudioEngine` stops and releases hardware wakelocks when the app enters `applicationDidEnterBackground`.	Resolve battery drain issues and normalize session duration metrics from 4,400s down to realistic ~180s averages.
P0	Monetization	**Verify sandbox payment receipts.** Audit the backend validation pipeline to ensure sandbox/TestFlight receipts are not being processed as production transactions.	Ensure accurate financial reporting and prevent fraudulent credit generation.
P1	Analytics	**Implement core feature telemetry.** Add custom GA4 event tracking for the main user actions: `clean_file_started`, `clean_file_completed`, `cleaning_error`, and `minutes_exhausted`.	Enable funnel tracking to understand feature usage and pinpoint drop-offs.
P2	Growth	**Migrate the app from a standalone title to a feature inside Music Separation.** Given its low user base (33 lifetime users) and high infrastructure costs, it makes strategic sense to bundle this utility as a premium feature inside the Music Separation iOS app.	Instantly expose the vocal cleaning engine to over 4,200 engaged users and lower overall maintenance overhead.

■ Visual Graphs & Performance Charts

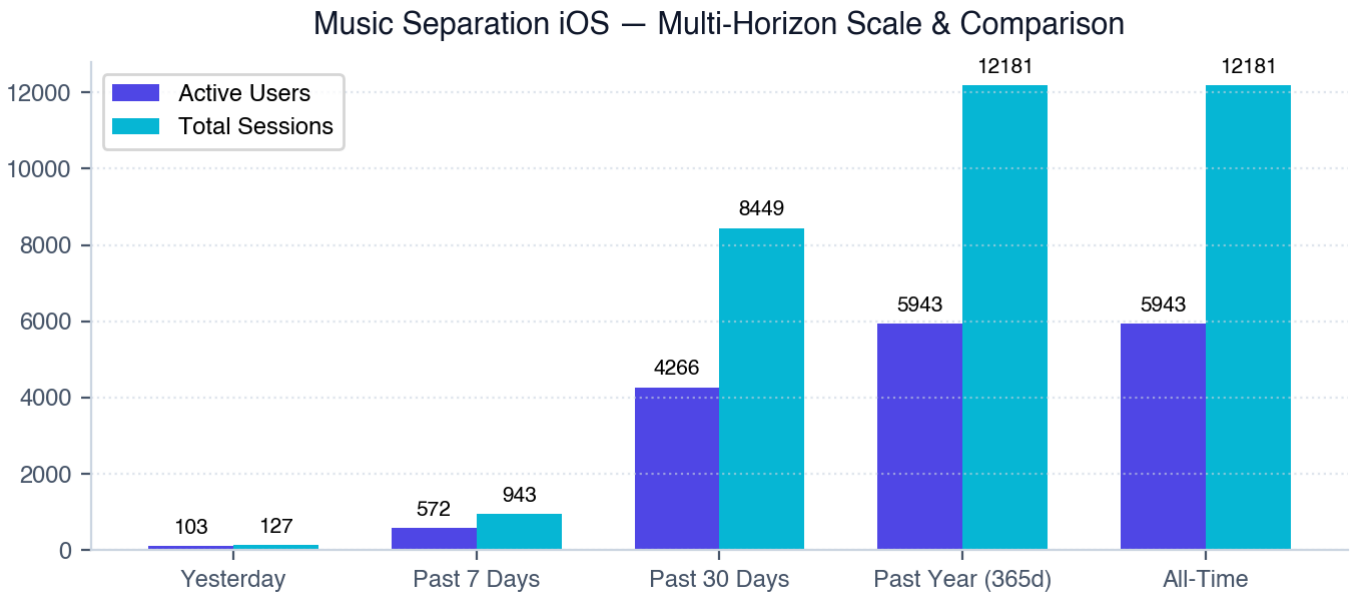
Music Separation Ios Trend



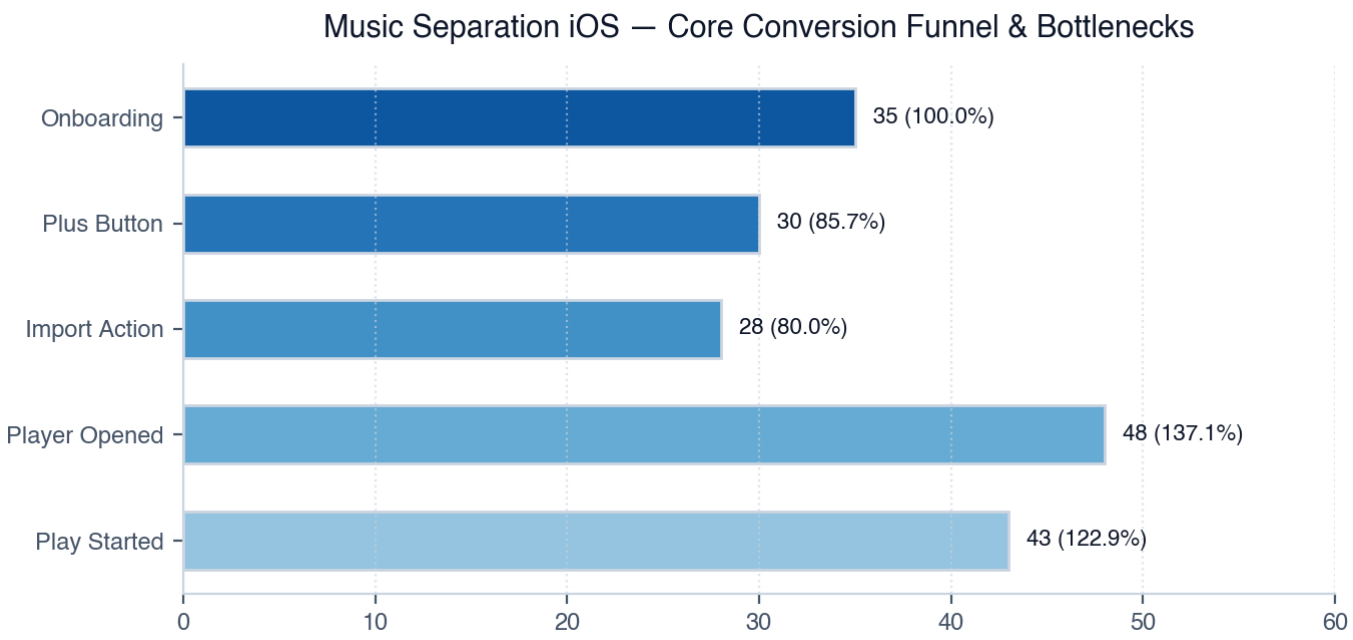
Music Separation Ios Revenue



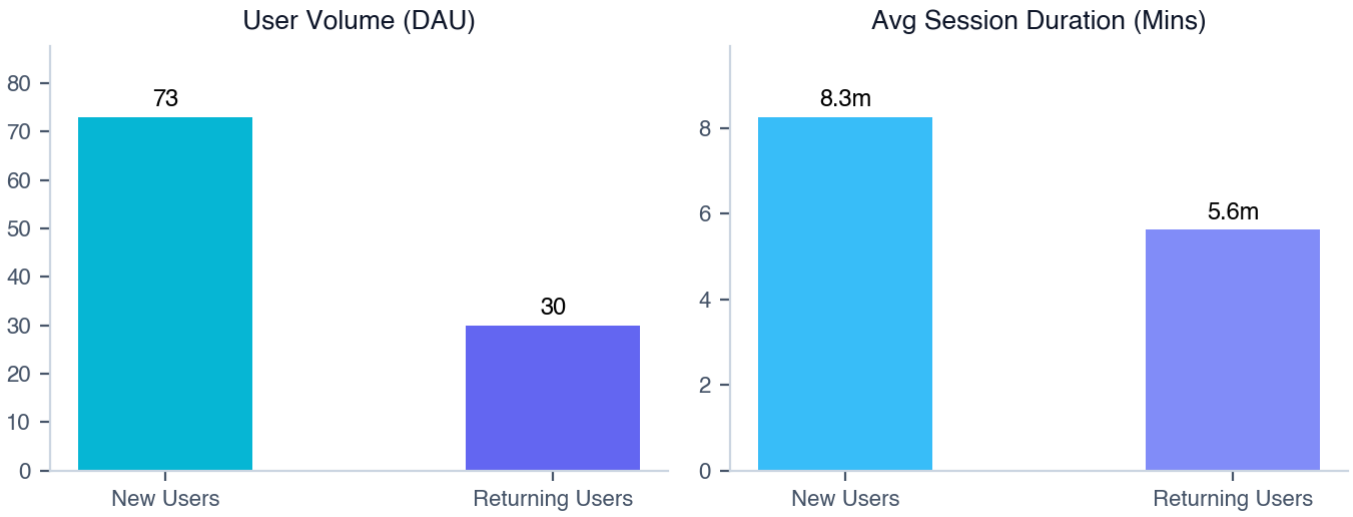
Music Separation Ios Horizons



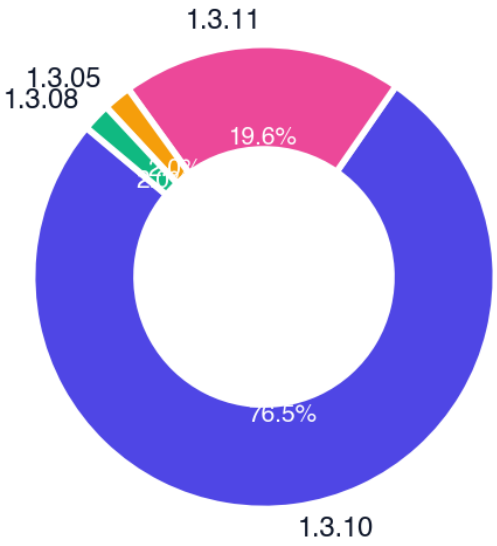
Music Separation Ios Funnel



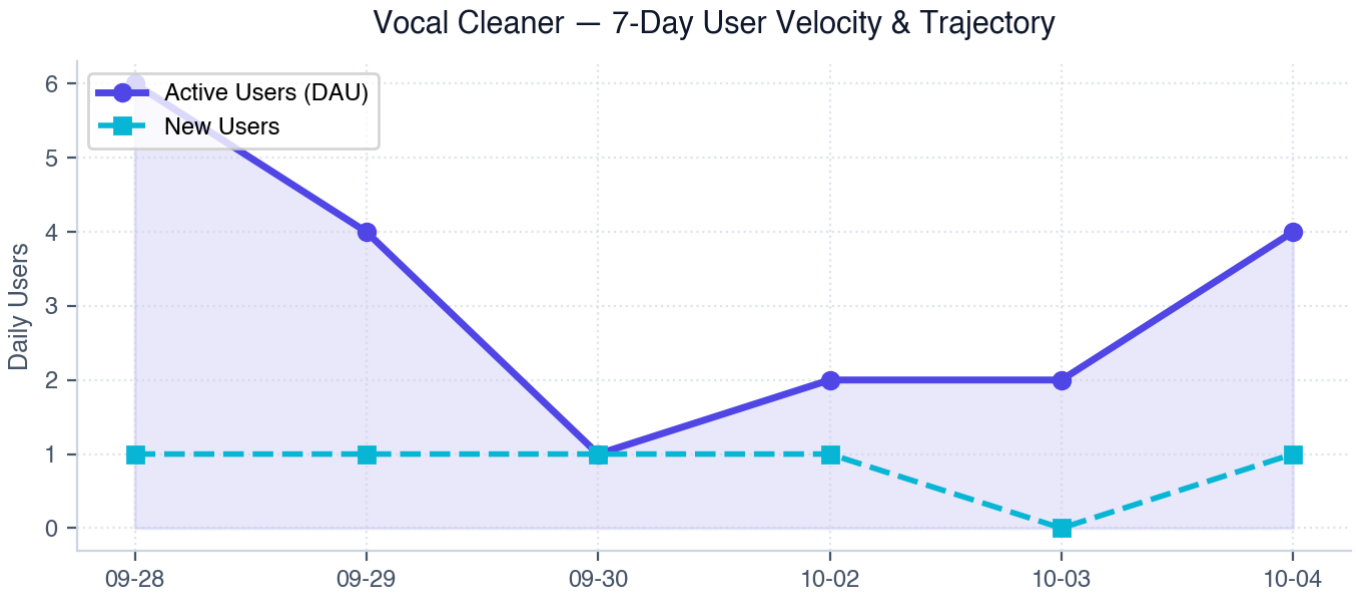
Music Separation Ios Cohort



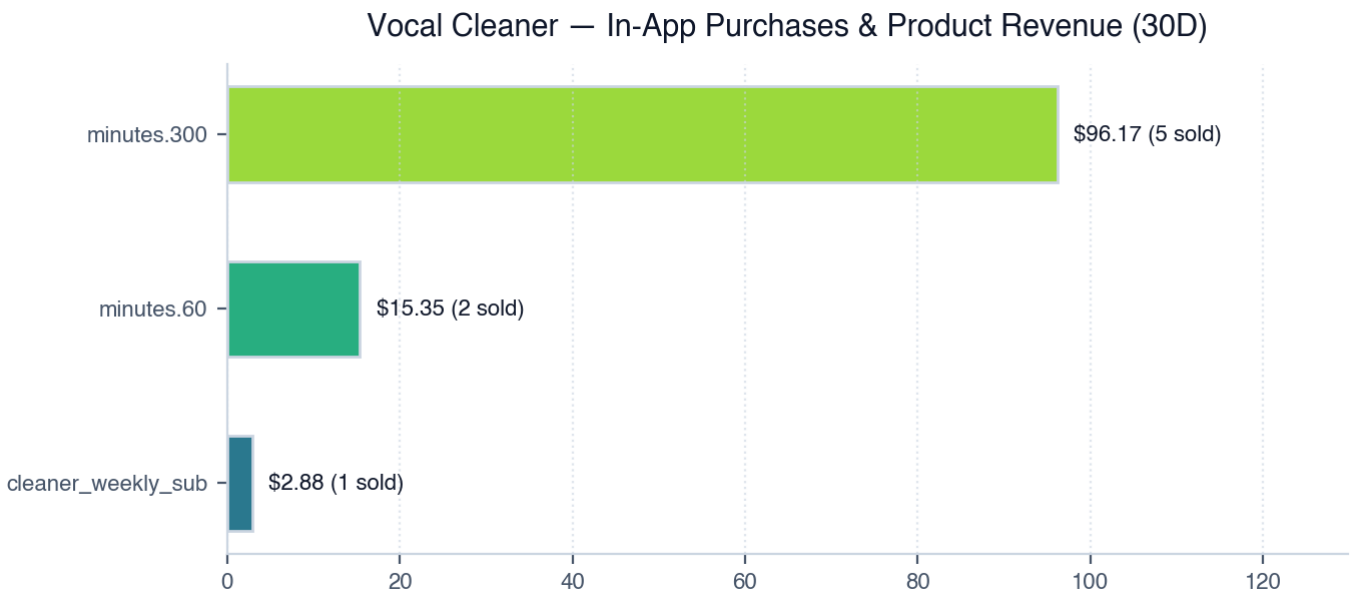
Music Separation Ios Version



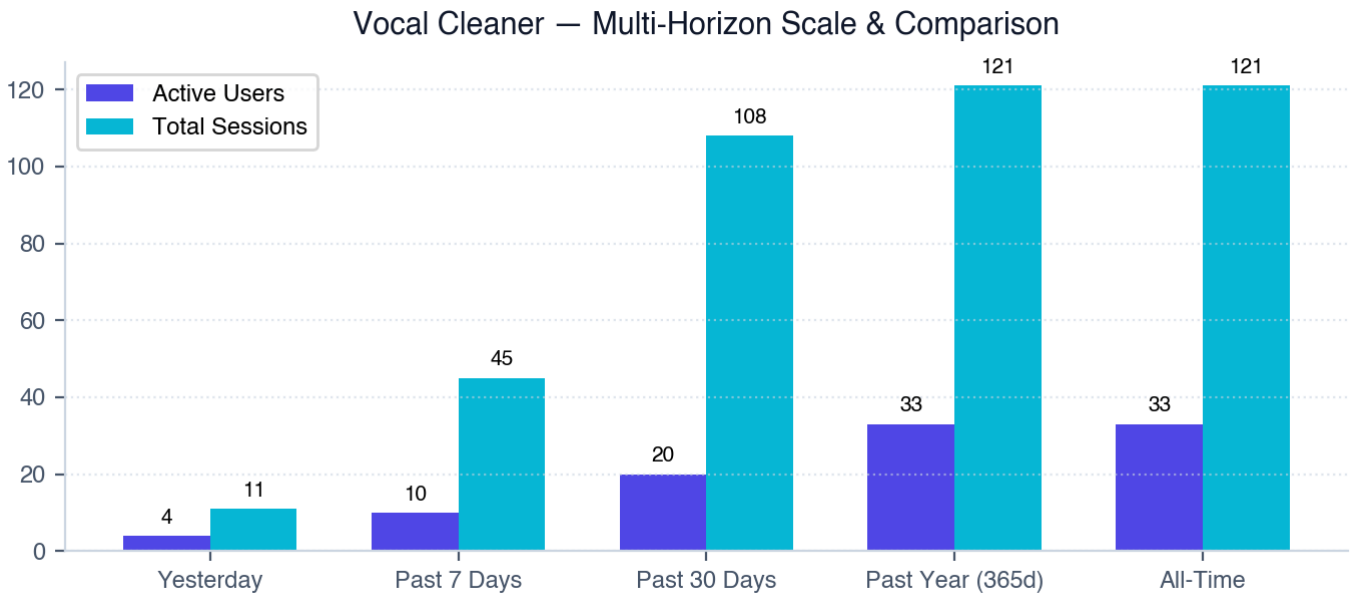
Vocal Cleaner Trend



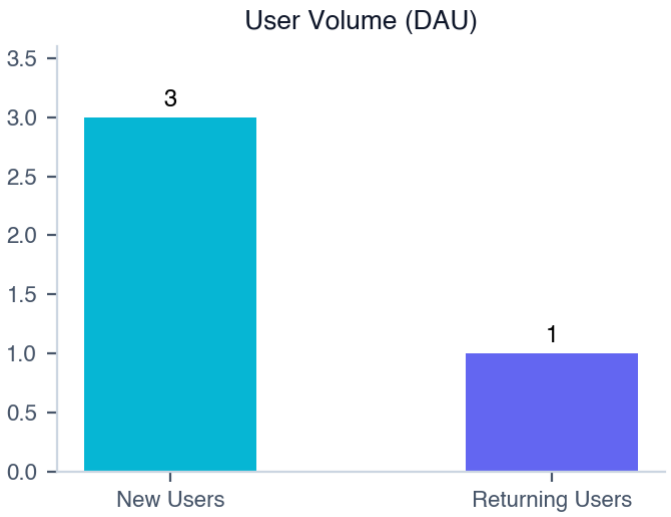
Vocal Cleaner Revenue

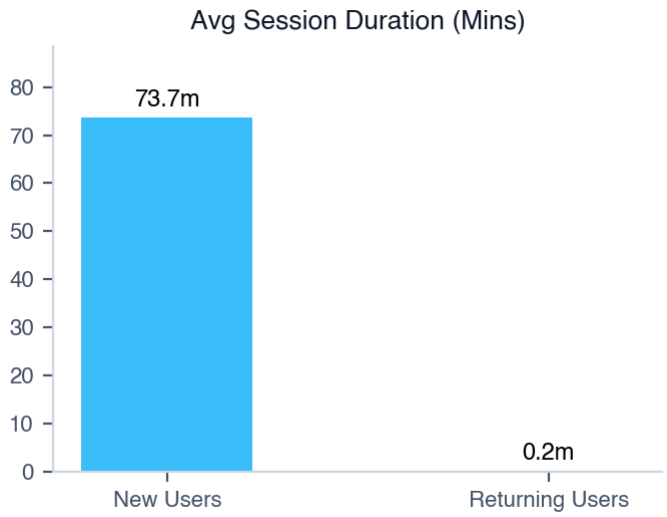


Vocal Cleaner Horizons

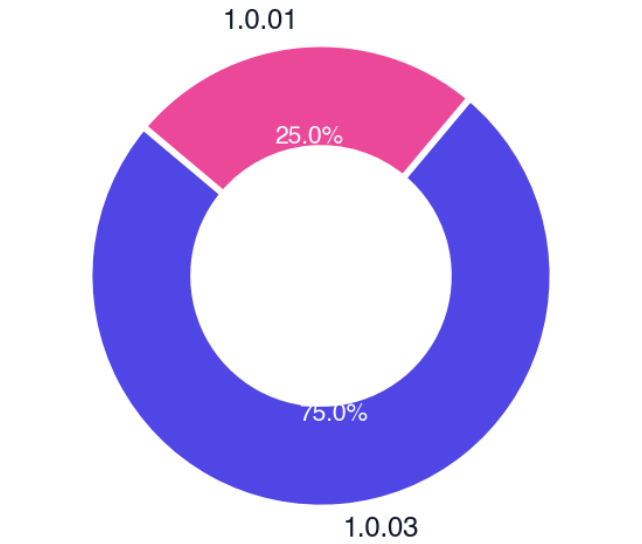


Vocal Cleaner Cohort





Vocal Cleaner Version



Direct Console Links:

■ Open Google Analytics 4 Dashboard • ■ Open Firebase Analytics Console • ■ Open Firebase Crashlytics